

Fake Job Detection

```
from google.colab import files
files.upload()
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

```
data = pd.read_csv("fake_job_postings.csv")

print(data.shape)
print(data["fraudulent"].value_counts()) # 0 = real, 1 = fake
data.head()
```

```
(17880, 18)
fraudulent
0    17014
1      866
Name: count, dtype: int64
```

	job_id	title	location	department	salary_range	company_profile	description	requirements	benefits
0	1	Marketing Intern	US, NY, New York	Marketing	NaN	We're Food52, and we've created a groundbreaki...	Food52, a fast-growing, James Beard Award-winn...	Experience with content management systems a m...	N
1	2	Customer Service - Cloud Video Production	NZ, , Auckland	Success	NaN	90 Seconds, the worlds Cloud Video Production ...	Organised - Focused - Vibrant - Awesome!Do you...	What we expect from you:Your key responsibilit...	What y will fr usThrot being p (
2	3	Commissioning Machinery Assistant (CMA)	US, IA, Wever	NaN	NaN	Valor Services provides Workforce Solutions th...	Our client, located in Houston, is actively se...	Implement pre-commissioning and commissioning ...	N
3	4	Account Executive - Washington DC	US, DC, Washington	Sales	NaN	Our passion for improving quality of life thro...	THE COMPANY: ESRI - Environmental Systems Rese...	EDUCATION: Bachelor's or Master's in GIS, busi...	(cultur anyth corpor —we ha
4	5	Bill Review Manager	US, FL, Fort Worth	NaN	NaN	SpotSource Solutions LLC is a Global Human Cap...	JOB TITLE: Itemization Review ManagerLOCATION:...	QUALIFICATIONS:RN license in the State of Texa...	I Bene Offe

```
data = data.fillna("")

data["text"] = data["title"] + " " + data["description"] + " " + data["requirements"] + " " + data["benefits"]
```

```
X = data["text"]
y = data["fraudulent"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

```
tfidf = TfidfVectorizer(stop_words="english", max_features=5000)

X_train_num = tfidf.fit_transform(X_train)
X_test_num = tfidf.transform(X_test)
```

```
model = LogisticRegression(max_iter=1000, class_weight="balanced")
model.fit(X_train_num, y_train)
```

LogisticRegression ⓘ ?
 LogisticRegression(class_weight='balanced', max_iter=1000)

```
predictions = model.predict(X_test_num)

print("Accuracy:", accuracy_score(y_test, predictions))
print(classification_report(y_test, predictions))
```

```
Accuracy: 0.9572147651006712
      precision    recall  f1-score   support

      0       0.99      0.96      0.98       3403
      1       0.54      0.83      0.65        173

   accuracy          0.96       3576
  macro avg          0.76      0.90      0.81       3576
 weighted avg          0.97      0.96      0.96       3576
```

```
def check_job(text):
    number_form = tfidf.transform([text])
    result = model.predict(number_form)[0]

    if result == 1:
        print("FAKE job")
    else:
        print("REAL job")
```

```
job1 = "Earn $5000 per day from home. No experience required. Send your bank details immediately. Limited offer."
job2 = "Urgent hiring! Work from home and earn big money. Pay a small registration fee of $50 to start. Contact us on WhatsApp"
job3 = "We are looking for a Python developer with 3 years of experience in SQL and REST APIs. Full time role with health insurance"
job4 = "Marketing executive needed for an established company. Responsibilities include managing social media and reporting campaigns"

print("Example 1:")
check_job(job1)
print("\nExample 2:")
check_job(job2)
print("\nExample 3:")
check_job(job3)
print("\nExample 4:")
check_job(job4)
```

```
Example 1:
FAKE job

Example 2:
FAKE job

Example 3:
REAL job

Example 4:
REAL job
```

```
my_job = input("Paste a job description here: ")
check_job(my_job)
```

```
Paste a job description here: We're Food52, and we've created a groundbreaking and award-winning cooking site. We support, connect, and inspire home chefs everywhere.
REAL job
```

```
my_job = input("Paste a job description here: ")
check_job(my_job)
```

```
Paste a job description here: The group has raised a fund for the purchase of homes in the Southeast. The student on this project is a member of the group.
FAKE job
```

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